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Ionic Elastomer Interface for Low-Pressure and Durable All-Solid-State Batteries

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ABSTRACT

All-solid-state batteries (ASSBs) hold significant promise as next-generation energy storage systems due to their high energy density and intrinsic safety. However, their practical deployment is impeded by the need for high external pressure (typically tens of megapascals) to maintain solid–solid interfacial contact and ensure long-term cycling stability. Here, we report an ionic elastomer specifically designed to enable stable ASSB operation under substantially reduced stack pressure. The elastomer combines a mechanically flexible polymer matrix with an ionically conductive phase, delivering high room-temperature ionic conductivity (0.3 mS cm^{-1}), a low elastic modulus (29.8 MPa), excellent thermal stability (≥400°C), and strong chemical compatibility with sulfide-based solid-state electrolytes (SSEs). When integrated with sulfide SSEs, the composite exhibits a remarkable room-temperature ionic conductivity of 5.23 mS cm^{-1}. Incorporation of this composite into ASSBs with high-nickel cathodes (Ni ≥ 90%) yields an initial capacity of 200 mAh g^{-1} at 0.05C and outstanding cycling stability over 700 cycles at 1C under a low stack pressure of just 5 MPa. This ionic-elastomer strategy mitigates electrochemical-mechanical degradation, eliminates the high-pressure requirement, and offers a scalable pathway toward practical, durable ASSBs technologies.

1 | Introduction

All-solid-state batteries (ASSBs) are among the most promising next-generation energy storage technologies, offering enhanced safety, high energy density, and compatibility with high-voltage, high-capacity electrode materials [1–3]. By replacing flammable liquid electrolytes with solid-state electrolytes (SSEs), ASSBs eliminate leakage risks and significantly improve operational safety [4–7]. However, a persistent challenge hindering their practical deployment is the need for high external pressure—typically tens of megapascals—to maintain intimate solid–solid interfacial contact and stable ion transport over extended cycling [8–13]. This pressure requirement imposes substantial barriers to commercial adoption, particularly in scenarios such as electric

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vehicles and grid-scale energy storage. The requirement for high stack pressure complicates system design, accelerates mechanical degradation, and limits form factor adaptability [14–16]. These challenges are exacerbated when using brittle sulfide-based SSEs paired with high-capacity electrodes, where repeated volume changes during cycling accelerate interfacial electrochemical–mechanical degradation under low-stack-pressure conditions. For instance, a $LiNi_{0.83}Co_{0.11}Mn_{0.06}O_2$ (NCM83)-based cell operated at 2 MPa retained only 65% of its initial capacity after 50 cycles, accompanied by a marked increase in interfacial voids revealed through 3D morphological reconstruction [17].

To overcome these challenges, several strategies have been developed. A notable example includes the use of a $Li_{21}Si_5$/Si–$Li_{21}Si_5$ double-layered anode to establish uniform electric fields and stable interfaces, enabling $LiCoO_2$-based cells to achieve 138.9 mAh g^{-1} and 80% capacity retention over 183 cycles without external pressure [18]. Similarly, incorporating an F-rich cathode interlayer and a $Mg_{16}Bi_{84}$ anode interlayer has been shown to mitigate internal stress, allowing $LiNiO_2$-based cells to deliver 11.1 mAh cm^{-2} and a cell-level energy density of 310 Wh kg^{-1} at just 2.5 MPa [19]. However, many of these systems require elevated temperatures (45°C–80°C) to obtain high performance, which limits their practicality for widespread applications [20–22]. Therefore, developing a general and scalable strategy to maintain stable solid–solid interfaces under low stack pressures is crucial for the practical applications of ASSB technology.

In this work, an ionically conductive elastomer is developed by combining a thermoplastic elastomer matrix with solvated ionic liquids. This material delivers high room-temperature ionic conductivity, excellent flexibility, outstanding thermal stability, and strong chemical compatibility with sulfide-based solid-state electrolytes. When integrated with sulfide SSEs, the resulting composite achieves an ionic conductivity of 5.23 mS cm^{-1} at room temperature. Leveraging this interfacial design, $LiNi_{0.9}Co_{0.09}Mo_{0.01}O_2$ (Ni90)–based ASSBs deliver an initial capacity of 200 mAh g^{-1} at 0.05 C and maintain stable cycling for 700 cycles at 1 C under an external pressure of only 5 MPa. This approach alleviates the high-pressure constraint inherent to ASSBs and mitigates electrochemical-mechanical degradation, offering a scalable route toward practical, durable solid-state battery technologies.

2 | Results and Discussion

2.1 | Schematic Illustration of the Role of the Ionic Elastomer in ASSBs

The thermoplastic elastomer poly(styrene-isobutylene-styrene) (SIBS) has been widely used as a base polymer for anion-exchange membranes due to its good thermal stability, chemical resistance, and mechanical robustness [23–25]. Structurally, SIBS consists of rigid polystyrene blocks at both ends and a soft polyisobutylene mid-block, which combines the processability of thermoplastics with the elasticity of rubber [26]. Compared with commonly used polymers such as PVDF or PEO, the nonpolar backbone of SIBS minimizes undesirable chemical interactions with sulfide electrolytes. These characteristics make SIBS an attractive mechanically compliant binder for ASSBs, provided that sufficient ionic conductivity can be introduced without compromising mechanical integrity. Building on this concept, we integrate the SIBS matrix with an ionically conductive phase composed of lithium bis(trifluoromethanesulfonyl)imide and triethylene glycol dimethyl ether. The resulting composite—referred to as SIBS-LiG3—exhibits high ionic conductivity at room temperature (3×10^{-4} S cm^{-1}), excellent mechanical resilience, and strong chemical compatibility with sulfide-based SSEs.

As illustrated in Figure 1a, insufficient external pressure during cycling leads to volumetric changes in the active materials, causing interfacial debonding between the cathode active material (CAM) and SSE [27–29]. Although conventional binders help preserve electrode integrity, their electronically and ionically insulating nature limits Li^+ transport, thereby compromising capacity [30]. In contrast, the incorporation of the ionically conductive SIBS-LiG3 elastomer simultaneously enhances interfacial adhesion and lithium-ion transport kinetics (Figure 1b). As a result, ASSBs incorporating this elastomeric binder could demonstrate desirable electrochemical performance even under reduced external pressure.

2.2 | Physicochemical Characterization of the Ionic Elastomer

The physicochemical properties of the ionic elastomer are systematically evaluated. To confirm the composition of SIBS, Raman spectroscopy was first employed. The characteristic peaks at 1000 and 2957 cm^{-1} corresponding to benzene ring stretching and C–H stretching vibrations, respectively, are attributed to the styrene and isobutylene segments [31, 32] (Figure S1). The optical image of SIBS before and after folding demonstrated its superior flexibility, as shown at the top of Figure 2a. After being endowed with Li^+ conductivity, the designed SIBS-LiG3 turns milky white while retaining its flexibility, as shown at the bottom of Figure 2a. The stress–strain curves of SIBS and SIBS-LiG3 were further characterized using a mechanical testing machine (Figure 2b; Figure S2). SIBS and SIBS-LiG3 exhibit moduli of 60.9 and 29.8 MPa, respectively, which indicates that the addition of Li-G3 maintains the mechanical strength of SIBS. Then the specific ionic conductivity of SIBS and LiG3 at different mass ratios (1:9, 3:7, and 5:5) was evaluated (Figure 2c; Figure S3), exhibiting conductivities of 0.31, 0.30, and 0.25 mS/cm, respectively. To balance high ionic conductivity and mechanical stability, a mass ratio of 3:7 was selected for further study and is referred to as SIBS-LiG3 hereafter. Consequently, the superior mechanical properties and high ionic conductivity of the SIBS-LiG3 provide a robust foundation for stable ion transport within sulfide-based solid-state batteries.

The structure and micromorphology of SIBS and SIBS-LiG3 were further analyzed. As shown in Figure 2d, Figure S4, SIBS exhibits a dense, cloth-like morphology. In contrast, SIBS-LiG3 (Figure 2e) forms an elastomeric network where the SIBS polymers are cross-linked, and the LiG3 phase is embedded within the elastomer matrix. In detail, the distribution of carbon elements outlines the elastomer skeleton's micromorphology (Figure 2f), while the LiG3 phase, identified by the presence of sulfur, is uniformly distributed within the elastomer framework (Figure 2g). This structure ensures that the LiG3 phase is well encapsulated

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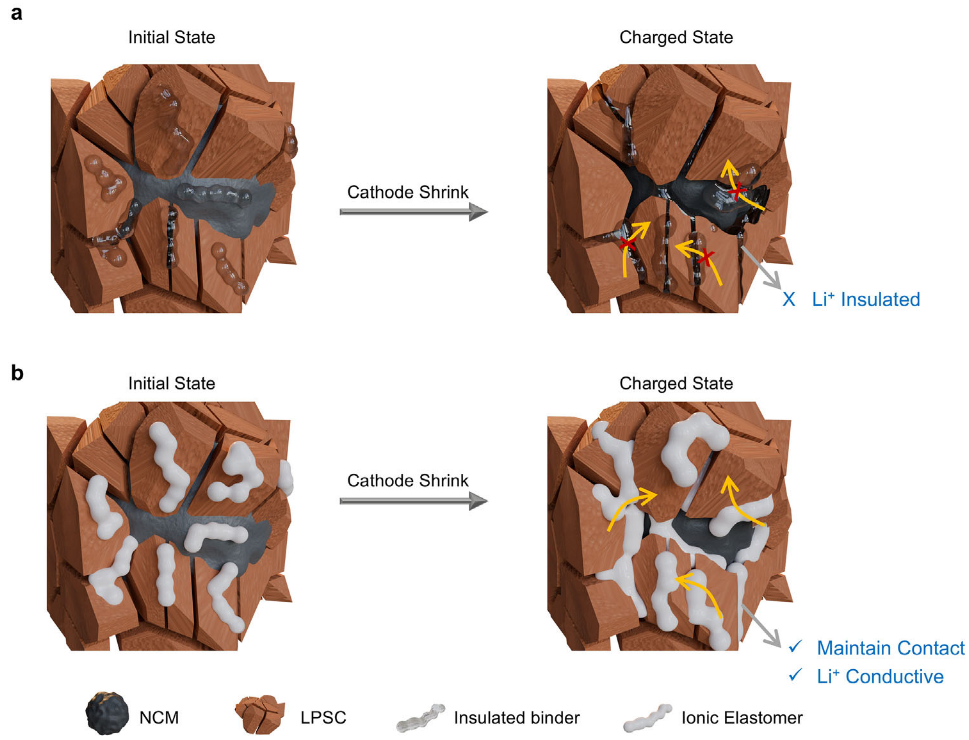

FIGURE 1 | (a) Insufficient pressure leads to interfacial detachment between CAM and SSE due to volume changes, while conventional binders hinder ion transport. (b) The ionic elastomer maintains interfacial contact and enables efficient Li^+ transport, supporting stable cycling under low pressure.

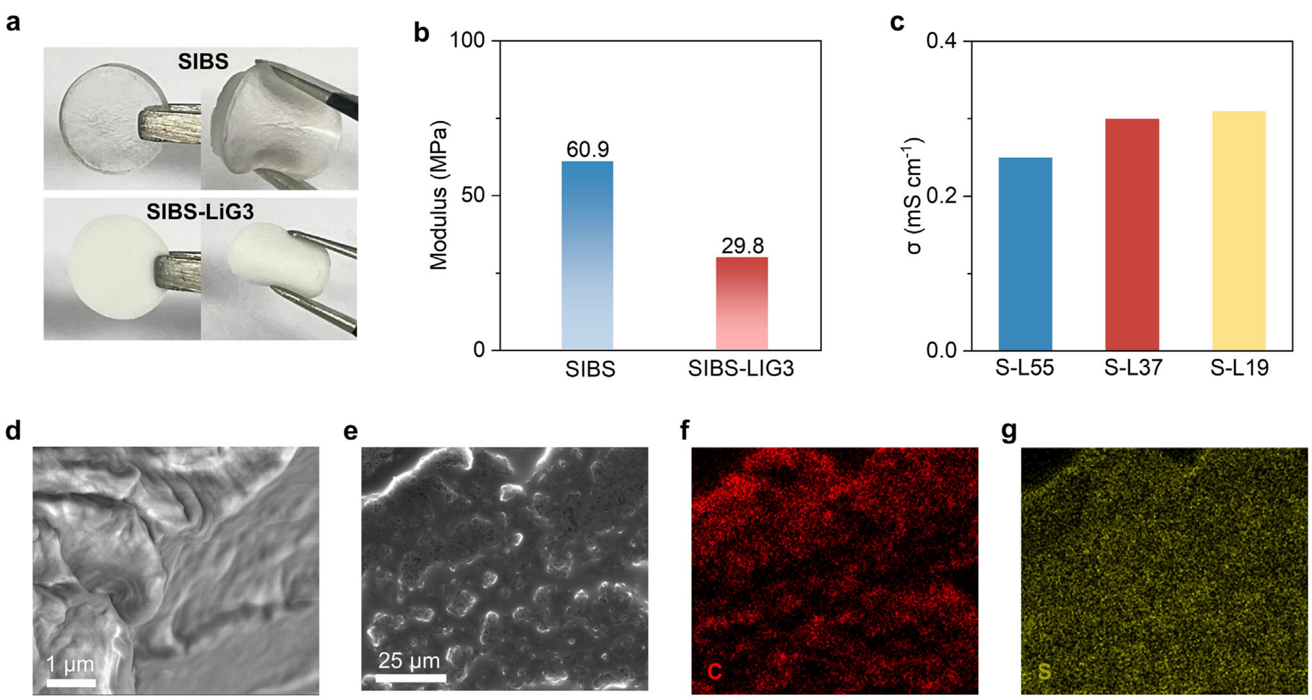

FIGURE 2 | Structural, mechanical, and electrochemical properties of the SIBS-based ionic elastomer. (a) Optical image demonstrating the flexibility of SIBS and SIBS-LiG3. (b) Elastic modulus measured via indentation tests. (c) Ionic conductivity of SIBS-LiG3 composites with varying weight ratios. (d,e) SEM images of pristine SIBS (d) and SIBS-LiG3 elastomer (e). (f,g) Elemental mapping of the SIBS-LiG3 composite shown in (e).

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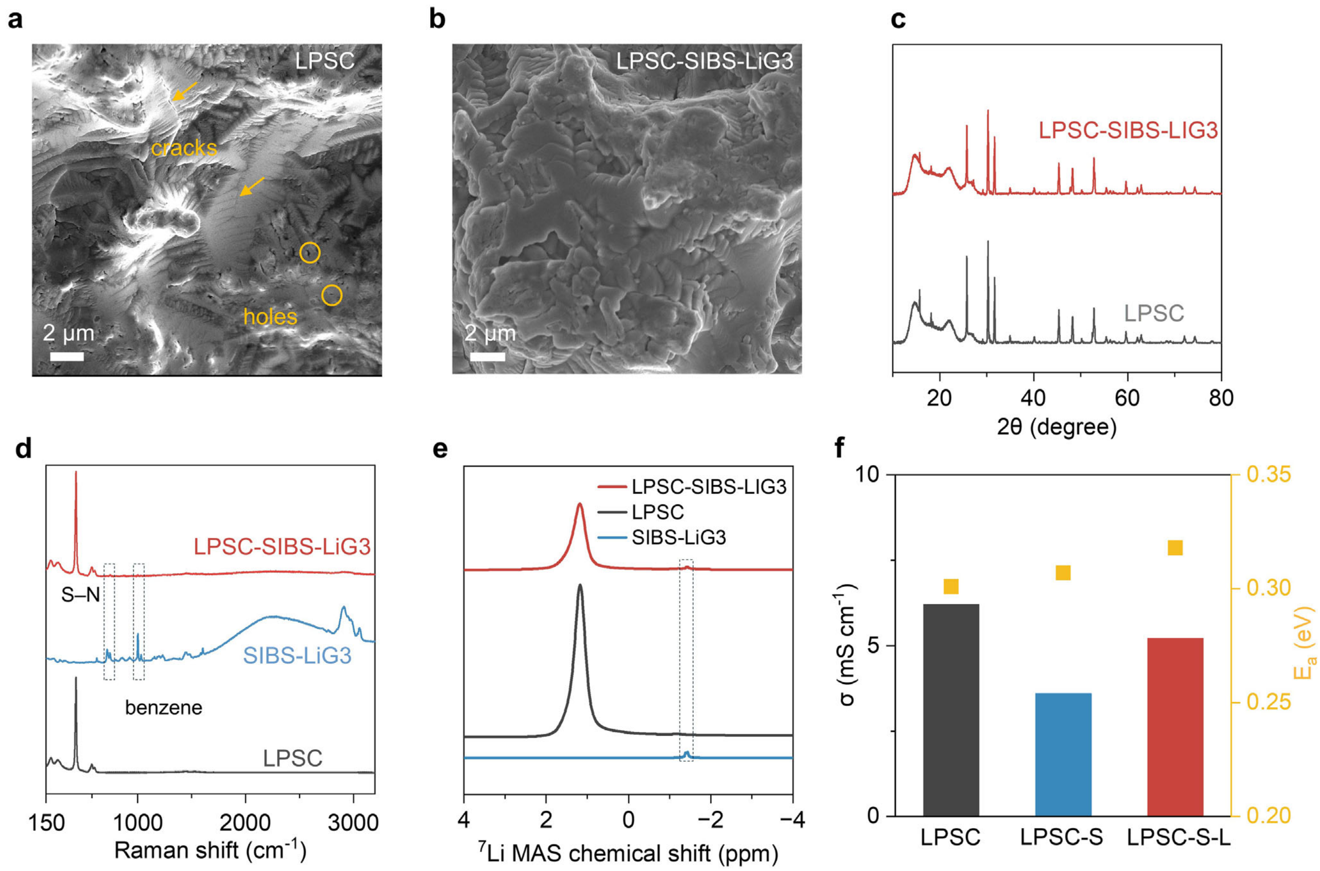

FIGURE 3 | (a) SEM image of a densified $Li_{5.5}PS_{4.5}Cl_{1.5}$ (LPSC) pellet. (b) SEM image of LPSC mixed with the ionic elastomer (LPSC-SIBS-LiG3). (c) XRD patterns of pristine LPSC and LPSC-SIBS-LiG3, indicating phase structural stability. (d) Raman spectra of LPSC, SIBS-LiG3, and their composite, showing no new phase formation. (e) MAS ^{7}Li NMR spectra of LPSC, SIBS-LiG3, and LPSC-SIBS-LiG3, confirming maintained lithium environments. (f) Ionic conductivity of LPSC, LPSC with SIBS, and LPSC with SIBS-LiG3, showing minimal conductivity loss with the ionic elastomer.

by the cross-linked SIBS, facilitating the formation of a 3D interconnected ionic network that serves as an efficient ion-conducting pathway. To evaluate thermal stability, differential scanning calorimetry (DSC) was performed (Figure S5). The thermogravimetric analysis of SIBS and SIBS-LiG3 revealed identical endothermic peaks extending up to 400°C under an inert atmosphere, confirming high thermal stability. Overall, these results demonstrate that incorporating the ion-conductive phase into the elastomer enhances its ionic conductivity while maintaining mechanical robustness and thermal stability.

2.3 | Compatibility of Ionic Elastomer With Sulfide-Based Solid-State Electrolytes

The compatibility and chemical stability between SIBS-LiG3 and LPSC were thoroughly evaluated. Dibromomethane (DBM), which is chemically stable with LPSC [33], was used as the solvent to prepare the LPSC-SIBS-LiG3 composite. As shown in Figures S6 and S7, the LPSC soaked in DBM for 24 h still maintains a pure phase and the same ion conductivity (~6.2 mS cm^{-1}), indicating excellent compatibility between DBM and LPSC. Then the composite sample of LPSC-SIBS-LIG3 (mass ratio 95:5) was prepared with the DBM. As shown in Figure 3a and Figure S8, the cross-section SEM of LPSC and LPSC-SIBS-LIG3 pellet densified at 250 MPa exhibits significantly different morphology. The rigid LPSC displays sharper edges, and much more cracks and holes due to the friction-induced stress concentration. In contrast, this phenomenon is suppressed in LPSC-SIBS-LIG3 due to the addition of ion-conducting elastomer (Figure 3b). As confirmed by SEM-EDS (Figure S9), the dark regions show a clear absence of sulfur signals, consistent with mechanical cracks or voids rather than compositionally distinct phases. By contrast, in the LPSC–SIBS–LiG3 composite (Figure S10), some dark regions are enriched with carbon signals, indicating that the SIBS elastomer partially fills or bridges these regions, thereby mitigating crack formation and preserving interparticle contact. This originates from the reduction of interparticle friction, which promotes the particle arrangement, leading to microstructural changes during the initial stages of densification [6, 34]. This enhanced particle packing subsequently promotes a more uniform plastic flow during densification, which improves the crack resistance property.

The phase components of LPSC and LPSC-SIBS-LiG3 were compared in Figure 3c, no other phase was observed except for the argyrodite (PDF#97-041-8490), which showed good chemical stability between LPSC and SIBS-LiG3 [35]. Raman spectrum measurements were further conducted to probe the SIBS-LiG3 in LPSC (Figure 3d). The peaks that emerged at 440 and 1005 cm^{-1} are attributed to S–N and benzene stretching, respectively, which exist in both the SIBS-LiG3 and LPSC-SIBS-LiG3 [32, 36].

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This further points toward the good compatibility between SIBS-LiG3 and LPSC. In addition, magic angle spinning (MAS) 7Li NMR measurements were carried out to determine the local environment around Li. As shown in Figure 3e, the 7Li MAS NMR spectrum of SIBS–LiG3 exhibits a single peak at approximately −1.4 ppm. This indicates that LiG3 is well incorporated into the SIBS matrix, resulting in a single and uniform ion-conductive elastomer. Additionally, two distinct peaks were observed in the ^{7}Li MAS NMR spectrum of LPSC–SIBS–LiG3 at 1.15 and −1.4 ppm, corresponding to LPSC and SIBS–LiG3, respectively, with no additional detectable signals [37]. This further confirms the absence of side reactions between LPSC and SIBS–LiG3. As a result, the LPSC-SIBS-LiG3 displays a high ion conductivity of 5.23 mS/cm at 25°C and an activation energy of 0.318 eV (Figure 3f; Figure S11). In contrast, without constructing the ion-conducting networks, the LPSC-SIBS displays an ion conductivity of 3.6 mS/cm due to the addition of ion insulated SIBS. Additionally, the LPSC-SIBS-LiG3 display a high thermal stability, which maintain the ion conductivity even exposed in 100°C for 24 h, Figure S12. Overall, these results demonstrate the enhanced crack resistance property and superior ion conductivity of LPSC-SIBS-LiG3.

2.4 | Electrode Dynamic Analysis Under Low Pressure

Encouraged by the perfect mechanical properties and ion conductivity, LPSC-SIBS-LiG3 is added to the sheet-type cathode. The micromorphology and compositions are first analyzed. As shown in Figure S13, SEM-EDS analysis of the non-densified cathode composite reveals that the SIBS-LiG3 phase is not homogeneously distributed as a bulk component, but preferentially locates at particle boundaries and NCM/LPSC interfacial regions, where contact loss is most likely to occur during cycling-induced volume changes. In contrast, the sheet-type cathode exhibits a uniform component distribution after densification (Figure S14). XRD patterns of NCM-SIBS-LiG3 show no detectable phase changes compared with pristine NCM (Figure S15), confirming that the slurry processing does not alter the cathode crystal structure. Raman spectra display the unchanged of characteristic peaks from both NCM and SIBS-LiG3, indicating good chemical compatibility between components.

To verify its improved electrode dynamics at low pressure, the electrochemical impedance spectroscopy (EIS) of the NCMo90-SIBS-based and NCMo90-SIBS-LiG3-based cell was measured at various operation pressures after applying a fabrication pressure of 250 MPa for pressure-induced densification, as shown in Figure 4a,b. Without external pressure (close to 0 MPa), both the cells show high total resistance mostly due to the insufficient contact between the cell and current collector. After applying a pressure of 5 MPa, the resistance of the cells decreased significantly. The NCMo90-SIBS-LiG3-based cell displays a smaller resistance (26.6 ohm) compared with the NCMo90-SIBS-based cell (38.7 ohm) due to the more sufficient ion-conducting networks at low pressure. As pressure increases to 100 MPa, the NCMo90-SIBS-LiG3-based cell displays a similar resistance compared to 5 MPa, indicating that applying over 5 MPa is enough to maintain the inside ion contact. The enhanced transport kinetics for Ni90-SIBS-LiG3 can also be supported by the dQ/dV profiles. As shown in Figure 4c, the obvious lower overpotential is observed for the phase transition process of Ni90-SIBS-LIG3. An overpotential of 30 mV is observed for Ni90-SIBS-LIG3 at H2/H3 phase change, but the NCMo90-SIBS displays an overpotential of 90 mV. Galvanostatic intermittent titration techniques (GITT) are further applied to confirm the Li^+ transport kinetics. As shown in Figure 4d, the lowered polarization for the Ni90-SIBS-LiG3 is observed in the whole charge and discharge process. Typically, as shown in Figure 4e, transient charge-discharge voltage profiles at 100% SOC show a lower overpotential of 40.6 mV for Ni90-SIBS-LiG3 than that of Ni90-SIBS for 64.5 mV, indicating a faster ion transport. The interfacial stability and dynamics of the LPSC-SIBS-LiG3 electrolyte against lithium metal was further evaluated using symmetric Li/LPSC-SIBS-LiG3/Li cells at 2 MPa. As shown in Figure S16, the composite electrolyte exhibits a lower interfacial resistance and a higher critical current density (CCD) than pristine LPSC, indicating improved interfacial stability and dynamics between Li metal and the composite electrolyte.

2.5 | Electrochemical Performance of ASSBs Under Low Pressure

Based on the improved low-pressure dynamic by SIBS-LiG3, the Ni90-SIBS-LiG3 cathode, LPSC-SIBS-LiG3 electrolyte and Li-In anode were assembled to evaluate the electrochemical performance of ASSBs at low pressure. As shown in Figure 5a, the Ni90-SIBS-LiG3-based cells display a specific capacity of 201.9 mAh/g and a Coulombic efficiency ~81.8% for the first cycle at 0.05 C at 5 MPa with a high mass loading of 12 mg cm^{-2}. After activated for 3 cycles, the cell cycles steadily at 0.1C for over 50 cycles with a coulombic efficiency of ~100%. The rate performance of Ni90-SIBS-LiG3 was also evaluated at 0.1C, 0.2C, 0.5C, 1C and 2C at 5 MPa, which exhibit a specific capacity of 192.2 178.2, 156.6, 143.3 and 110.1 mAh/g, respectively (Figure 5b). The corresponding voltage curves in Figure 5c show the stable charge and discharge process without soft short-circuit at high current density. The superior rate performance at low pressure is attributed to the enhanced dynamics provided by SIBS-LIG3. In contrast, the Ni90-SIBS only exhibits a specific capacity of 165.9, 155.9, 138.5, 126.3 and 89.7 mAh/g, respectively, due to the slow dynamic (Figure S11). Furthermore, the long-term cyclability of Ni90-SIBS-LiG3 cells at low pressure was confirmed (Figure 5d). Excitingly, the Ni90-SIBS-LiG3 displays an initial capacity of 127.9 mAh g^{-1} at 1C and shows a capacity of 98.1 mAh g^{-1} after 700 cycles, surpassing the performance of representative low-pressure ASSBs reported previously (Figure 5g). Meanwhile, the Ni90-SIBS-LiG3 displays an initial capacity of 148.9 mAh g^{-1} at 0.5C with a capacity retention of 81% over 450 cycles. (Figure S12). Additionally, the cross-section SEM of Ni90-SIBS-LiG3 cycled for 10th displays a similar morphology to the initial state (Figure S13), indicating that few electrochemical-mechanical degradation has occurred. This behavior is attributed to the limited stress-induced damage, as evidenced by the small stress change observed in Figure 5f. The cell exhibits small normalized stress variations of 0.27, 0.23, and 0.19 MPa mAh^{-1} at 0.1C, 0.2C, and 0.5C, respectively, which is also qualitatively comparable to reported all-solid-state batteries using Li-In alloy anodes [45]. To isolate the stress evolution originating from the cathode, $Li_4Ti_5O_{12}$ (LTO) was employed as the counter electrode owing to its negligible volume change (~0.2%) during lithiation and delithiation (Figure S20). Under this configuration, a characteristic stress change of ~0.13 MPa mAh^{-1} is obtained. In

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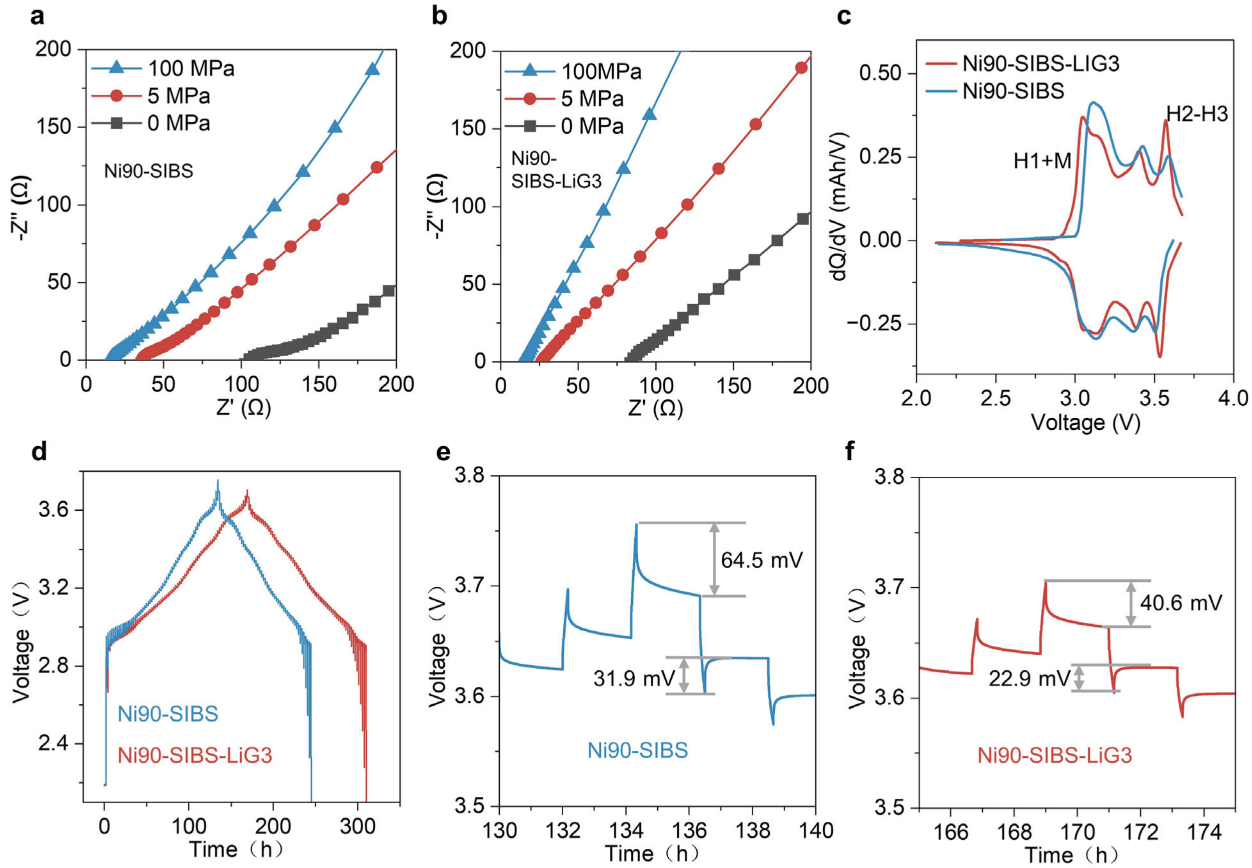

FIGURE 4 | (a,b) Electrochemical impedance spectra (EIS) of Ni90-SIBS and Ni90-SIBS-LiG3 cells at varying external pressures. (c) Differential capacity (dQ/dV) curves comparing Ni90-SIBS and Ni90-SIBS-LiG3 cells. (d) GITT profiles of both cells during the first cycle. (e,f) Enlarged potential responses during the lithium insertion/extraction steps in (d), highlighting Li^+ transport kinetics.

addition, to demonstrate the practicality, an all-solid-state pouch cell was further fabricated and tested under lower operation pressure. As shown in Figure 5d, the pouch cell achieves a capacity of 130.9 mAh g^{-1} at 0.5C (5 mAh) and cycled steadily for over 50 cycles at 2 MPa. Then, the cycled pouch cell was unloaded and operated free from external pressure (<0.1 MPa), as shown in Figure S14, the cell still achieves a capacity of 102.8 mAh g^{-1} at 0.5C and cycled steadily for over 1000 cycles. To assess performance under more practical condition, full cells were assembled using (i) a low-lithium-content In-based anode (N/P ≈ 1.3) and (ii) a pure In anode (anode-free configuration). As shown in Figure S22, both configurations maintain stable cycling, although reduced capacity is observed, which can be attributed to sluggish Li^+ transport and partial lithium trapping in In metal. Post-cycling SEM and EDS analyses after 50 cycles (Figures S23 and S24) reveal that the cathode composite retains a relatively uniform morphology without pronounced cracking or delamination. These results suggest that the LPSC-SIBS-LiG3 electrolyte helps preserve interfacial contact and mitigates mechanical degradation even under limited lithium supply.

3 | Conclusion

In this work, we present an ionically conductive elastomer that exhibits high room-temperature ionic conductivity, mechanical flexibility, thermal stability, and excellent compatibility with sulfide SSEs. The ionic elastomer, composed of a thermoplastic elastomer matrix and solvated ionic liquids, effectively binds the inorganic components together, maintaining interfacial contact and facilitating ionic transport during cycling under low pressure. The slurry-prepared composite of sulfide SSEs and the ionic elastomer achieves an ionic conductivity of 5.23 mS cm^{-1} at room temperature. Leveraging this composite, ASSBs with high-nickle cathodes deliver a high specific capacity of 200 mAh g^{-1} at 0.05C and sustain over 700 stable cycles at 1C under a low external pressure of 5 MPa at 25°C. This strategy markedly lowers the mechanical constraints traditionally associated with high-energy ASSBs, mitigates electrochemical–mechanical degradation, and paves the way for their practical commercialization.

4 | Experimental Section

4.1 | Preparation of the Materials

LiG3 was synthesized by blending equimolar amounts of LiTFSI (lithium bis(trifluoromethanesulfonimide, 99.9%, Sigma–Aldrich) and triglyme (G3, 99%, Sigma–Aldrich). SIBS-LiG3 composites were prepared by mixing SIBS (Kaneka Corporation) with LiG3 at weight ratios of 1:9, 3:7, or 5:5 in dibromomethane (DBM, 99.9%, Sigma–Aldrich) as the solvent. The Ni90-based cathode was composed of commercial $LiNi_{0.9}Co_{0.09}Mo_{0.01}O_2$ and $Li_{5.5}PS_{4.5}C_{1.5}$ solid electrolyte in a 70:30 (wt.%) ratio. The mixture

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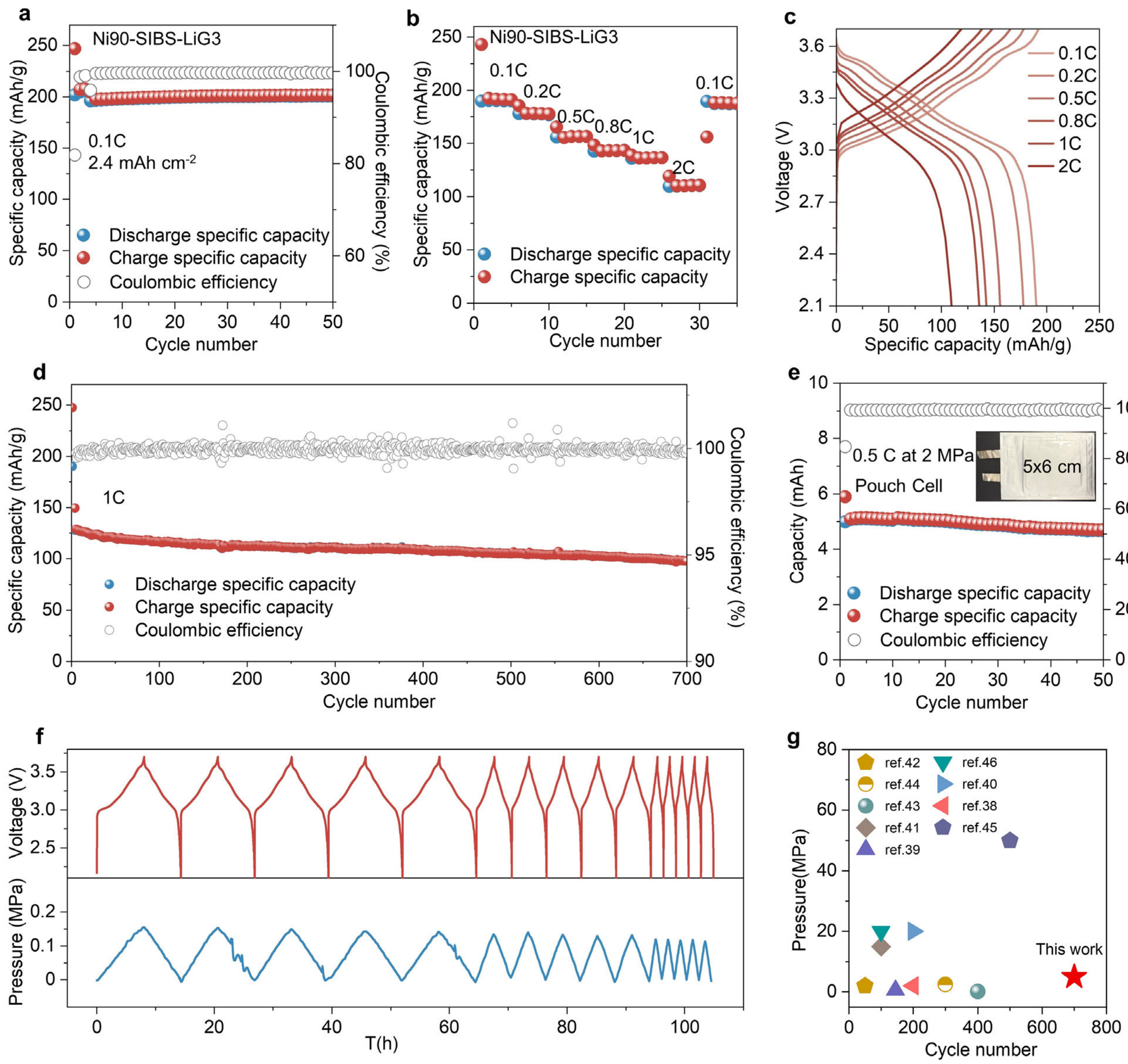

FIGURE 5 | Electrochemical performance of Ni90-SIBS-LiG3 cells under low external pressure. (a) Cycling stability of the Ni90-SIBS-LiG3 cell at 5 MPa and 0.1C. (b) Rate capability of the Ni90-SIBS-LiG3 cell at 5 MPa. (c) Charge-discharge profiles at various current rates. (d) Long-term cycling performance at 1C and 5 MPa. (e) Cycling performance of a pouch cell at 2 MPa. (f) In situ stress response of Ni90-SIBS-LiG3 cells during charge/discharge at different rates. (g) Comparison of long-term cycling stability at low pressure with previously reported data, highlighting the superior performance of this work [15, 38–44].

was loaded into a zirconia jar with zirconia balls and ball-milled at 300 rpm for 2 h using a planetary milling apparatus.

4.2 | Preparation of the Cathode and Solid Electrolytes

Ni90-SIBS-LiG3 (or Ni90-SIBS) cathode slurries were prepared by dispersing Ni90-based active material and SIBS-LiG3 (or SIBS) in a 95:5 (wt.%) ratio in DBM. The resulting slurry was coated onto carbon-coated aluminum current collectors using a doctor blade, followed by vacuum drying at 80°C for 12 h. The LPSC–SIBS-LiG3 solid electrolyte composite was similarly prepared by mixing LPSC and SIBS-LiG3 at a 95:5 (wt.%) ratio in DBM to form a slurry. All preparation steps were performed in an argon-filled glovebox (H_2O < 0.1 ppm, O_2 < 0.1 ppm) to prevent moisture or air contamination.

4.3 | Cell Fabrication

Mold-type solid-state cells were assembled via cold pressing using the following steps: a) 80 mg of LPSC–SIBS-LiG3 was added to a mold and pressed at 125 MPa (1 ton) for 1 min to form a flat electrolyte layer. b) A Ni90-SIBS-LiG3 (or Ni90-SIBS) electrode slice (4–5 mg) was placed onto the electrolyte and pressed at 250 MPa (2 tons) for 3 min. c) A 30 µm indium film was placed on the opposite side of the electrolyte layer, followed by a 20 µm lithium foil on top of the In film. d) The four-layer pellet was pressed at 125 MPa (1 ton) for 1 min. The fully assembled pellet

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was sandwiched between two stainless-steel rods and sealed in a homemade mold cell.

For pouch cell assembly: The Ni90-SIBS-LiG3 cathode was prepared as described above. The anode was formed by placing lithium foil directly on an indium film. The solid electrolyte membrane was fabricated by mixing LPSC–SIBS-LiG3 with polytetrafluoroethylene (PTFE, GLABAT) at a 99.5:0.5 (wt.%) ratio using a Hummer Acoustic Mixer (HAM100). This dough was rolled into films using a hot roller (TMAXCN) at 60°C. Electrodes and electrolytes were cut to size using a pneumatic punch (Anode and electrolyte: 50 mm × 60 mm, cathode: 45 mm × 56 mm). Ultrasonic welding was used to connect tabs to current collectors. The components were stacked, sealed in a laminate pouch, and subjected to 300 MPa pressure for 90 s using a BTKF high-pressure press (Baotou Kefa High Pressure Technology Co., Ltd). The final cathode active material loading was ~57 mg. All assembly steps were performed in an argon-filled glovebox (H_2O < 0.1 ppm, O_2 < 0.1 ppm).

4.4 | Electrochemical Test

Galvanostatic charge–discharge tests were carried out using a battery testing system (LAND CT2001A) at 25°C. Electrochemical impedance spectroscopy (EIS) was conducted on a BioLogic VSP-300 electrochemical workstation over a frequency range of 7 MHz–20 mHz with an applied AC amplitude of 10 mV. Galvanostatic intermittent titration technique (GITT) measurements were performed at a rate of 0.1C with 10 min current pulses, followed by 2 h relaxation intervals to allow for voltage stabilization.

4.5 | Characterization

The phase composition of the materials was identified using powder X-ray diffraction (XRD; Rigaku). Microstructural analysis was performed using field-emission scanning electron microscopy (FESEM; Hitachi). Raman spectroscopy (LabRAM HR Evolution) was conducted with a 623.8 nm argon-ion laser. Solid-state ^{7}Li nuclear magnetic resonance (SSNMR) measurements were carried out on a Bruker AVANCE NEO 600 WB spectrometer (B_0 = 14.1 T, Larmor frequency ω_0 = 233.24 MHz for ^{7}Li) using a 3.2 mm HXY MAS probe at room temperature. Mechanical properties were evaluated using an indentation-based compression method (Iesttech SPFT-20) at 25°C with a constant displacement rate of 1 μm/S. Stress evolution during charge/discharge cycling was monitored using a custom-built pressure-measuring device integrated with a force sensor (CHUANGLI, Ningbo). The assembled cell was tested at 25°C under a constant applied pressure of 5 MPa, and all stress variations during cycling were recorded using proprietary software.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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